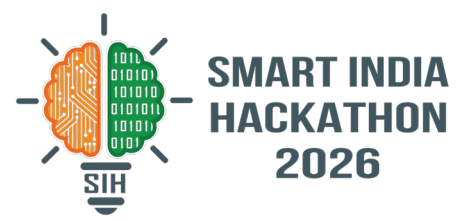


SMART INDIA HACKATHON 2026



VAJRA वज्र

Multi-source convective nowcasting with honest lead times

- Problem Statement ID – 26084
- Problem Statement Title – Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)
- Theme – Disaster Management
- PS Category – Software
- Team ID – 159951
- Team Name (Registered on portal) – CodeX_2026



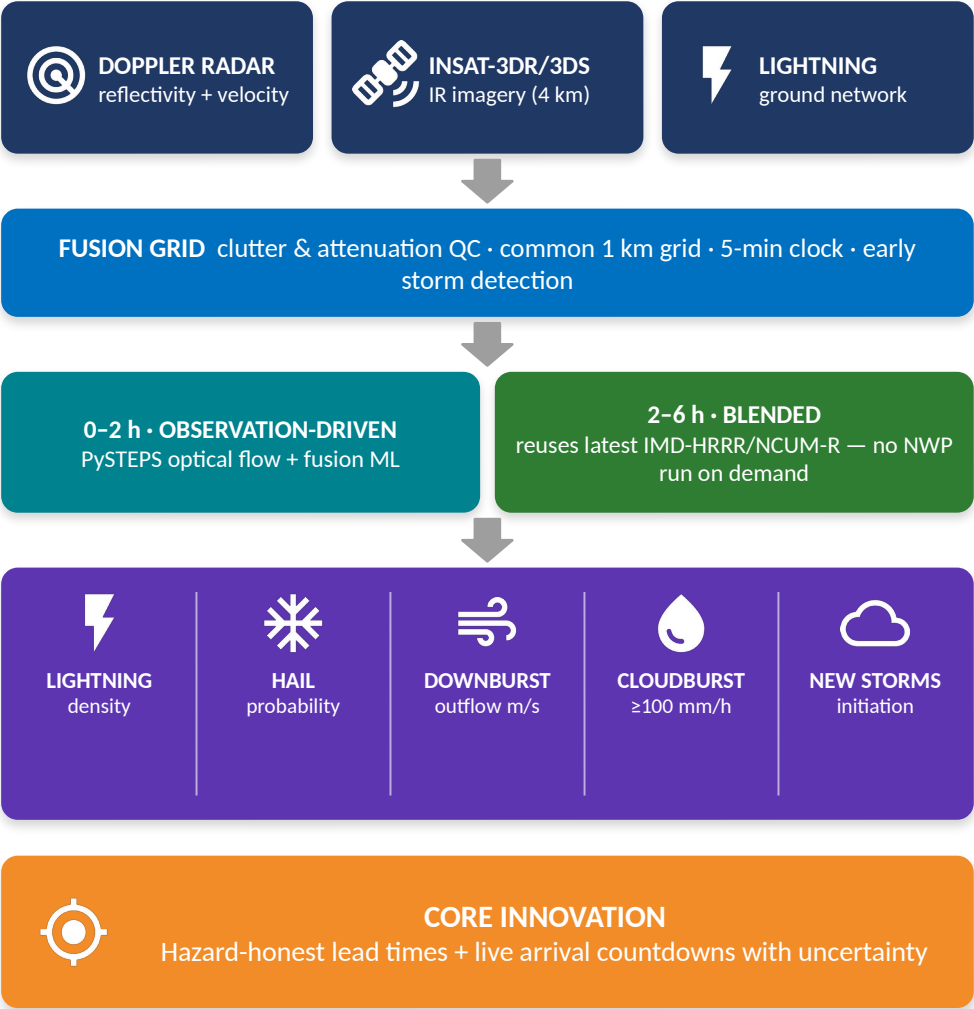
VAJRA: Fusing Radar, Satellite & Lightning into Honest 0–6 h Storm Nowcasts

Real-world issue: Thunderstorms, hail, downbursts and cloudbursts form in minutes over a few km — too fast and small for regular NWP forecasts.

Why important: IMD's radar nowcasts are strong for ~1–2 h, but hazard-specific, fused forecasts out to 6 h — and coverage outside radar — are missing.

Solution: Fuse DWR, INSAT IR and lightning; detect new storms early; forecast each hazard separately; blend with NWP to 6 h; live arrival countdowns.

[Demo · GitHub](#) ← **Prototype**



Problem	↔	Our Fix
Skill fades after ~2 h	↔	Blend with NWP to 6 h
Radar gaps, satellite 4 km	↔	Satellite early-storm detection
One-size warnings	↔	Separate forecast per hazard
Countdowns without uncertainty	↔	Live countdown + window + %

What exists vs. what's missing

Existing (IMD/IITM)	Gap	VAJRA
SWIRLS-2, PySTEPS	Mostly 0–2 h radar	Fused, to 6 h
DAMINI lightning	~40 min, one hazard	All hazards
IMD-HRRR	Not blended w/ obs	Skill-weighted blend

Sources: Atmosphere 15(2):154, 2024 (IMD nowcasting tools) · J. Earth Syst. Sci. 2025 · MOSDAC INSAT-3D payloads · IITM DAMINI

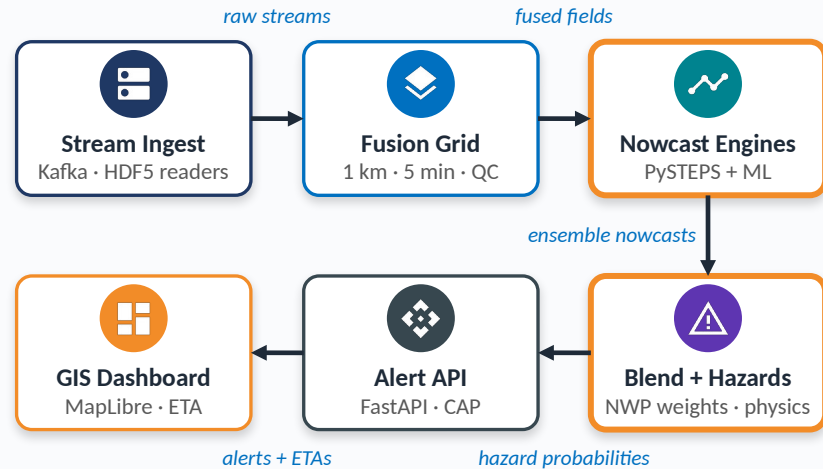
TECHNICAL APPROACH

METHODOLOGY & PROCESS OF IMPLEMENTATION

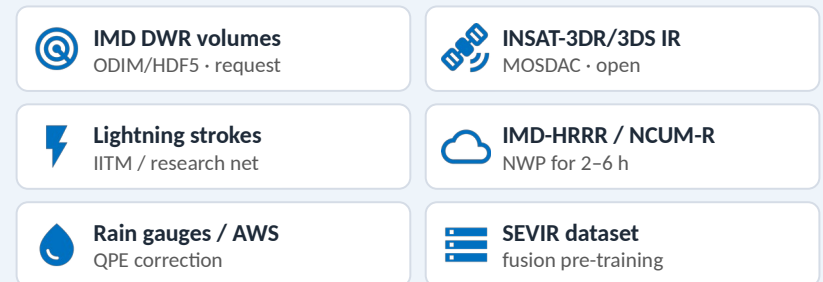


Graceful degradation: 1 km outputs under radar coverage; ~4 km satellite + lightning guidance elsewhere, clearly labelled.

SYSTEM ARCHITECTURE



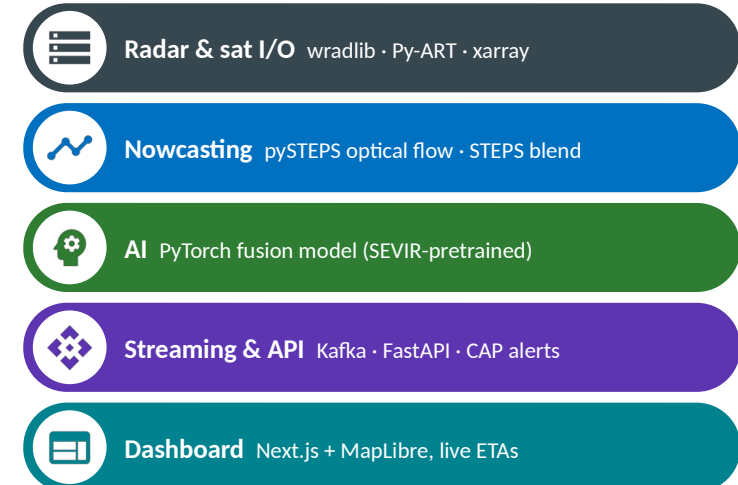
DATA STREAMS & TRAINING



KEY FEATURES

- ✓ Live countdown to each storm's arrival window (range + %)
- ✓ Downburst outflow speed from Doppler divergence
- ✓ Radar QC: clutter removal, attenuation correction
- ✓ Target: nowcast out ≤5 min after each radar scan
- ✓ Reuses latest NWP output — no model run on demand
- ✓ Alerts: CAP → SACHET, SMS/IVR, DAMINI, local languages

TECHNOLOGIES USED



Sources: pySTEPS (Pulkkinen et al., GMD 2019) · arXiv 2404.10512 (STEPS blending) · MOSDAC · IMD nowcasting tools (Atmosphere 2024)

**DATA**

INSAT open (MOSDAC) · SEVIR open · IMD radar on request

**TOOLS**

Open-source: pySTEPS, wradlib, PyTorch

**COMPUTE**

Nowcast in seconds; training on 1 GPU

**SCOPE**

Kolkata nor'westers + 1 cloudburst case

**TECHNICAL FEASIBILITY***Analysis of the feasibility of the idea***Proven building blocks**

Optical-flow nowcasting and STEPS blending are established; IMD already runs PySTEPS.

**Data exists**

INSAT via MOSDAC now; SEVIR offers co-located radar, satellite and lightning for training.

**Real-time capable**

Extrapolation + ML inference run in seconds; target: nowcast published ≤5 min after each radar scan.

**Measurable**

CSI, FSS, Brier score, reliability and ETA error against observations.

**RISKS & MITIGATION***Potential challenges · strategies for overcoming them*

Risk	Level	Mitigation strategy
No open Indian radar / lightning data	High	SEVIR pre-training; MOSDAC; formal request via SIH SPOC
Over-claiming 6 h skill	High	Per-hazard skill curves; NWP blending beyond ~2 h
Blurry ML forecasts	Med	Generative model; judged by FSS, not just average error
Satellite IR only 4 km	Med	1 km only under radar; labelled confidence elsewhere
Downburst lead is minutes	Med	Treated as a warning, not a 6 h forecast
Duplicating IMD tools	Low	PySTEPS as baseline; we add fusion + hazards

**VIABILITY & DEPLOYMENT***Path from pilot to operations***PHASE 1 · SIH**

Case replays: Kolkata nor'westers, 1 cloudburst

**PHASE 2 · Pilot**

Live with one IMD radar + INSAT + lightning

**PHASE 3 · National**

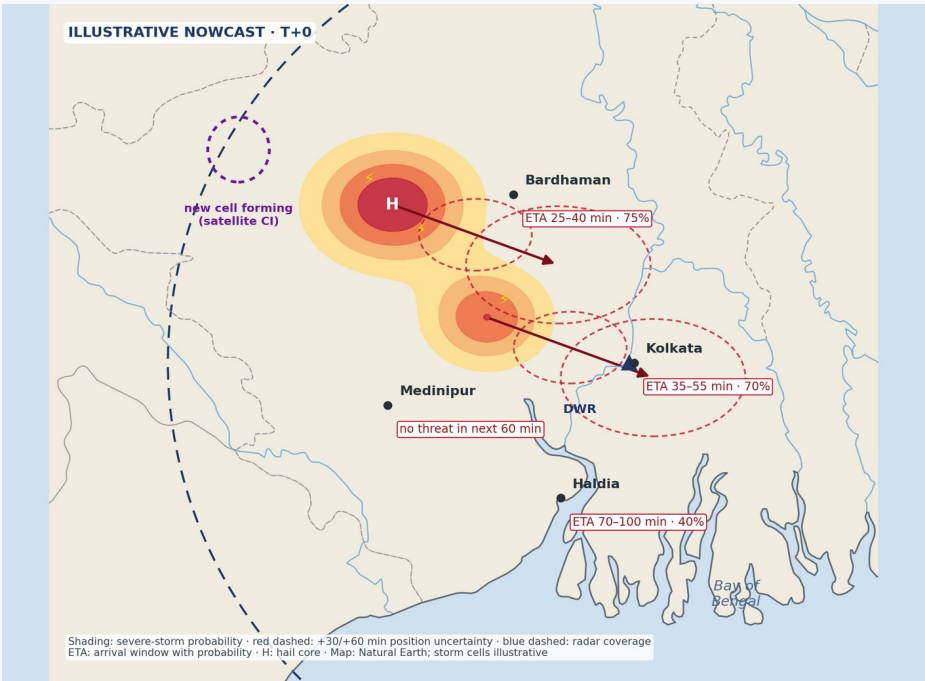
Mission Mausam radars; SACHET, SMS/IVR & DAMINI alerts

Why now: IMD plans village-level nowcasts every 10 minutes; Mission Mausam is expanding the radar network VAJRA builds on.

Owner: IMD nowcast desks use it · alerts via SACHET · open-source, MoES-owned

Sources: pySTEPS · SEVIR (MIT Lincoln Lab) · MOSDAC · IMD press release, Jan 2025 · J. Earth Syst. Sci. 2025

 WHAT A FORECASTER SEES: STORMS, HAZARDS, ARRIVAL WINDOWS
Illustrative nowcast on real geography (Natural Earth), Kolkata region



-  Lightning density
-  Hail core
-  Satellite: new storm forming
-  Live countdown: window + %

 HONEST LEAD TIMES, HAZARD BY HAZARD
Confidence by lead time · to be replaced with measured skill

Hazard	0-1 h	1-2 h	2-6 h
Lightning	High	Medium	Area probability
Hail	High (radar)	Medium	Environment risk
Downburst (m/s)	Minutes-30 min	Low	Not at point scale
Cloudburst	High (radar rain)	Medium	NWP-blended prob.
New storms	Satellite 15-60 min	Medium	NWP-driven

A dashboard that shows the same confidence for every hazard at 6 hours would be wrong — VAJRA shows what the science supports.

IMPACT ON TARGET AUDIENCE · BENEFITS: lives saved (social), avoided losses (economic), safer operations (aviation)

 **IMD nowcast desk**
One fused view of all hazards

 **Disaster managers**
Live countdowns to move people & teams

 **Aviation**
Hail & downburst zones on approach paths

 **Farmers (via IMD)**
SMS/IVR & DAMINI alerts in local languages

Sources: IMD cloudburst definition (100 mm/h over ~20-30 km²) · IITM DAMINI · IMD press release, Jan 2025 · lightning deaths estimate: IITM/Shaastra

RESEARCH AND REFERENCES

Details / Links of the reference and research work · every entry is clickable

IMD SYSTEMS	SATELLITE & LIGHTNING	METHODS	AI MODELS	DATA & DEFINITIONS	DETAILED REPORT
1 IMD radar nowcasting tools (WDSS-II, SWIRLS-2, PySTEPS) Atmosphere · 2024 LINK ↗ 2 SWIRLS nowcasting with Indian DWR MAUSAM LINK ↗ 3 Village-level 10-minute nowcast plan IMD press release · Jan 2025 LINK ↗	1 INSAT-3D imager bands (4 km IR) MOSDAC / ISRO LINK ↗ 2 INSAT-3DR rapid-scan imaging Current Science LINK ↗ 3 India's lightning networks IIT Madras Shastra LINK ↗	1 Radar nowcasting skill, NE India J. Earth Syst. Sci. · 2025 LINK ↗ 2 pySTEPS nowcasting library Pulkkinen et al. · GMD 2019 LINK ↗ 3 Satellite diffusion nowcasting; STEPS blending arXiv 2404.10512 LINK ↗	1 DGMR: skilful precipitation nowcasting Ravuri et al. · Nature 2021 LINK ↗ 2 NowcastNet: extreme precipitation nowcasting Zhang et al. · Nature 2023 LINK ↗ 3 Nowcast3D: 3D radar nowcasting arXiv 2511.04659 LINK ↗	1 SEVIR: radar + satellite + lightning MIT Lincoln Lab LINK ↗ 2 Cloudburst definition (IMD) IITM explainer LINK ↗	 Scan for: Detailed report Code & prototype <i>[Insert QR]</i>

KEY FINDINGS THAT SHAPED OUR DESIGN *what the research says → what we built because of it*

